

MICROSERVICES ARCHITECTURE



Structures an application as a collection of small, independent, loosely coupled services, each running in its own process and communicating via APIs or messaging.

KEY CHARACTERISTICS OF MICROSERVICES

Feature	Description
Decentralized	Each service owns its data and logic.
Independent Deployment	Services can be updated without redeploying the entire system.
Polyglot Persistence	Different services can use different databases (SQL, NoSQL).
Scalability	Services scale independently (e.g., scale only the payment service during peak sales).
Resilience	Failure in one service doesn't crash the whole system.

WHEN TO USE MICROSERVICES?

- ✓ Scalable cloud applications (e.g., Uber, Netflix)
- ✓ Teams working independently
- ✓ Need for polyglot persistence
- ✓ Frequent deployments

Challenges

- ✗ Complexity (service discovery, distributed transactions)
- ✗ Network latency (inter-service calls)
- ✗ Debugging difficulties

Next Steps:

- Add **Kubernetes** for orchestration
 - Implement **API Gateway (Ocelot, Kong)**
 - Use **GraphQL** for efficient data fetching
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AUTHENTICATION & AUTHORIZATION IN MICROSERVICES

Architecture Overview

Component	Responsibility
Auth Service	Issues JWT tokens, handles login/register
API Gateway	Validates tokens, routes requests
Other Services	Verify roles/permissions from JWT

Key Technologies

- JWT (JSON Web Tokens) - Stateless authentication
- OAuth2 - For federated identity (e.g., "Login with Google")
- Role-Based Access Control (RBAC) - patient, doctor, admin roles
- Spring Security (Java) or Passport.js (Node.js)

AUTHENTICATION & AUTHORIZATION IN MICROSERVICES

Role-Based Access Control (RBAC)

Define permissions in a centralized policy:

Role	Permissions
patient	Read own records, book appointments
doctor	Read/write patient records, prescribe meds
admin	Full access

[//Auth Service Implementation \(Node.js\)](#)

SECURITY BEST PRACTICES

BEST PRACTICES

- Use HTTPS everywhere
- Store secrets in environment variables/KMS
- Short-lived JWT tokens (15-30 min expiry)
- Refresh tokens for long-lived sessions
- Rate limiting to prevent brute force attacks
- Audit logs for all access attempts

HIPAA Compliance Considerations

- Encrypt PHI (Protected Health Information) in transit and at rest
 - Role minimum privilege (e.g., nurses shouldn't delete records)
 - Token revocation for compromised accounts
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